Secure Sight Full Evaluation Report

Total runs: 25

Total batches: 5

1. Overview

This full evaluation report includes all results found in the evaluation folder, including statistical analysis, global analysis, batch comparisons, and detailed run summaries.

2. Batch Comparison Summaries

Batch 1

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SecureSight Batch Comparison Report — batch_1
Generated: 2026-05-03T18:00:31.108659
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Total runs: 5

RUN SUMMARY:
    run_1 | Acc: 0.8958 | F1: 0.8928 | Loss: 0.1916 | Latency: 8.00ms
    run_2 | Acc: 0.8685 | F1: 0.8607 | Loss: 0.2267 | Latency: 8.07ms
    run_3 | Acc: 0.8916 | F1: 0.8819 | Loss: 0.2103 | Latency: 7.76ms
    run_4 | Acc: 0.8797 | F1: 0.8832 | Loss: 0.2122 | Latency: 7.88ms
    run_5 | Acc: 0.8958 | F1: 0.8928 | Loss: 0.1916 | Latency: 8.00ms

STABILITY METRICS:
  Accuracy: mean=0.8863 ± 0.0119
  F1:       mean=0.8823 ± 0.0131
  Loss:     mean=0.2065 ± 0.0150
  Latency:  mean=7.94 ± 0.12ms

Best run:  run_1 (F1=0.8928, Acc=0.8958)
Worst run: run_2 (F1=0.8607, Acc=0.8685)

HYPERPARAMETERS:
  learning_rate: 0.0001
  batch_size: 32
  dropout: 0.3
  optimizer: adamw
  weight_decay: 1e-05
  seed: 43

COMPARISON PLOTS:
  ✓ batch_runs_accuracy.png
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✓ batch_runs_f1.png
✓ batch_runs_loss.png
✓ batch_runs_latency.png
✓ batch_radar_comparison.png

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Batch 2

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SecureSight Batch Comparison Report — batch_2
Generated: 2026-05-03T18:04:14.459484
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Total runs: 5

RUN SUMMARY:
run_1 | Acc: 0.8485 | F1: 0.8407 | Loss: 0.2367 | Latency: 8.07ms
run_2 | Acc: 0.8716 | F1: 0.8619 | Loss: 0.2203 | Latency: 7.76ms
run_3 | Acc: 0.8597 | F1: 0.8632 | Loss: 0.2222 | Latency: 7.88ms
run_4 | Acc: 0.8758 | F1: 0.8728 | Loss: 0.2016 | Latency: 8.00ms
run_5 | Acc: 0.8485 | F1: 0.8407 | Loss: 0.2367 | Latency: 8.07ms

STABILITY METRICS:
Accuracy: mean=0.8608 ± 0.0127
F1: mean=0.8559 ± 0.0145
Loss: mean=0.2235 ± 0.0145
Latency: mean=7.96 ± 0.14ms

Best run: run_4 (F1=0.8728, Acc=0.8758)
Worst run: run_1 (F1=0.8407, Acc=0.8485)

HYPERPARAMETERS:
learning_rate: 5e-05
batch_size: 32
dropout: 0.2
optimizer: adamw
weight_decay: 1e-05
seed: 44

COMPARISON PLOTS:

✓ batch_runs_accuracy.png
✓ batch_runs_f1.png
✓ batch_runs_loss.png
✓ batch_runs_latency.png
✓ batch_radar_comparison.png

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Batch 3

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SecureSight Batch Comparison Report — batch_3
Generated: 2026-05-06T14:19:21.232776
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Total runs: 5

RUN SUMMARY:
run_1 | Acc: 0.9116 | F1: 0.9019 | Loss: 0.2003 | Latency: 9.31ms
run_2 | Acc: 0.9158 | F1: 0.9128 | Loss: 0.1816 | Latency: 9.60ms
run_3 | Acc: 0.8885 | F1: 0.8807 | Loss: 0.2167 | Latency: 9.68ms
run_4 | Acc: 0.9158 | F1: 0.9150 | Loss: 0.1934 | Latency: 9.86ms
run_5 | Acc: 0.8815 | F1: 0.8665 | Loss: 0.1715 | Latency: 9.72ms

STABILITY METRICS:
Accuracy: mean=0.9026 ± 0.0164
F1: mean=0.8954 ± 0.0211
Loss: mean=0.1927 ± 0.0174
Latency: mean=9.63 ± 0.21ms

Best run: run_4 (F1=0.9150, Acc=0.9158)
Worst run: run_5 (F1=0.8665, Acc=0.8815)

HYPERPARAMETERS:
learning_rate: 0.0002
batch_size: 64
dropout: 0.4
optimizer: adamw
weight_decay: 1e-05
seed: 53

COMPARISON PLOTS:
✓ batch_runs_accuracy.png
✓ batch_runs_f1.png
✓ batch_runs_loss.png
✓ batch_runs_latency.png
✓ batch_radar_comparison.png

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Batch 4

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SecureSight Batch Comparison Report — batch_4
Generated: 2026-05-03T18:10:39.198470
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Total runs: 5

RUN SUMMARY:
run_1 | Acc: 0.8858 | F1: 0.8828 | Loss: 0.1966 | Latency: 9.60ms
run_2 | Acc: 0.8585 | F1: 0.8507 | Loss: 0.2317 | Latency: 9.68ms
run_3 | Acc: 0.8816 | F1: 0.8719 | Loss: 0.2153 | Latency: 9.31ms
run_4 | Acc: 0.8858 | F1: 0.8850 | Loss: 0.2084 | Latency: 9.86ms
run_5 | Acc: 0.8515 | F1: 0.8365 | Loss: 0.1865 | Latency: 9.72ms

STABILITY METRICS:
Accuracy: mean=0.8726 ± 0.0164
F1: mean=0.8654 ± 0.0211
Loss: mean=0.2077 ± 0.0174
Latency: mean=9.63 ± 0.21ms

Best run: run_4 (F1=0.8850, Acc=0.8858)
Worst run: run_5 (F1=0.8365, Acc=0.8515)

HYPERPARAMETERS:
learning_rate: 0.0001
batch_size: 64
dropout: 0.1
optimizer: adamw
weight_decay: 1e-05
seed: 58

COMPARISON PLOTS:
✓ batch_runs_accuracy.png
✓ batch_runs_f1.png
✓ batch_runs_loss.png
✓ batch_runs_latency.png
✓ batch_radar_comparison.png

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Batch 5

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SecureSight Batch Comparison Report — batch_5
Generated: 2026-05-06T14:20:31.586260
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Total runs: 5

RUN SUMMARY:
run_1 | Acc: 0.8840 | F1: 0.8733 | Loss: 0.1689 | Latency: 11.69ms
run_2 | Acc: 0.8840 | F1: 0.8733 | Loss: 0.1689 | Latency: 11.69ms
run_3 | Acc: 0.9004 | F1: 0.8813 | Loss: 0.1987 | Latency: 12.01ms
run_4 | Acc: 0.9017 | F1: 0.8913 | Loss: 0.1859 | Latency: 12.09ms
run_5 | Acc: 0.8817 | F1: 0.8627 | Loss: 0.2280 | Latency: 11.97ms

STABILITY METRICS:
Accuracy: mean=0.8904 ± 0.0098
F1: mean=0.8764 ± 0.0106
Loss: mean=0.1901 ± 0.0246
Latency: mean=11.89 ± 0.18ms

Best run: run_4 (F1=0.8913, Acc=0.9017)
Worst run: run_5 (F1=0.8627, Acc=0.8817)

HYPERPARAMETERS:
learning_rate: 8e-05
batch_size: 128
dropout: 0.3
optimizer: adamw
weight_decay: 1e-05

seed: 63

COMPARISON PLOTS:

✓ batch_runs_accuracy.png

✓ batch_runs_f1.png

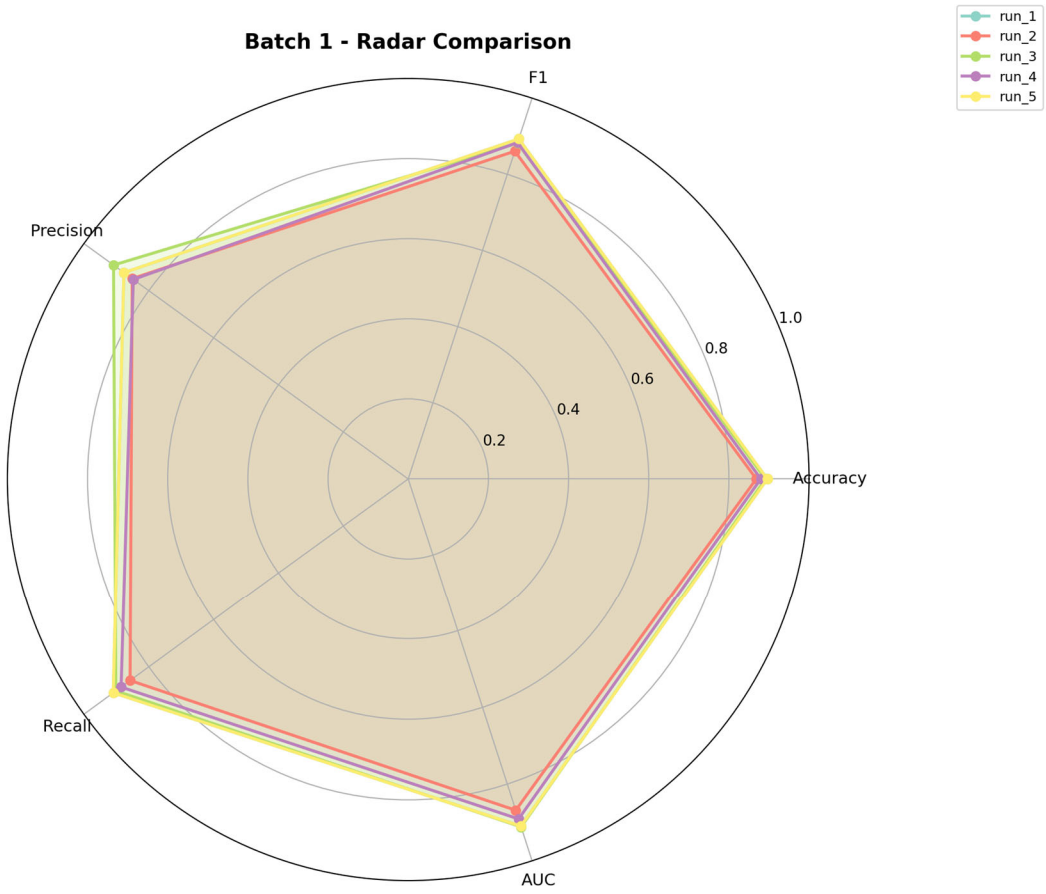
✓ batch_runs_loss.png

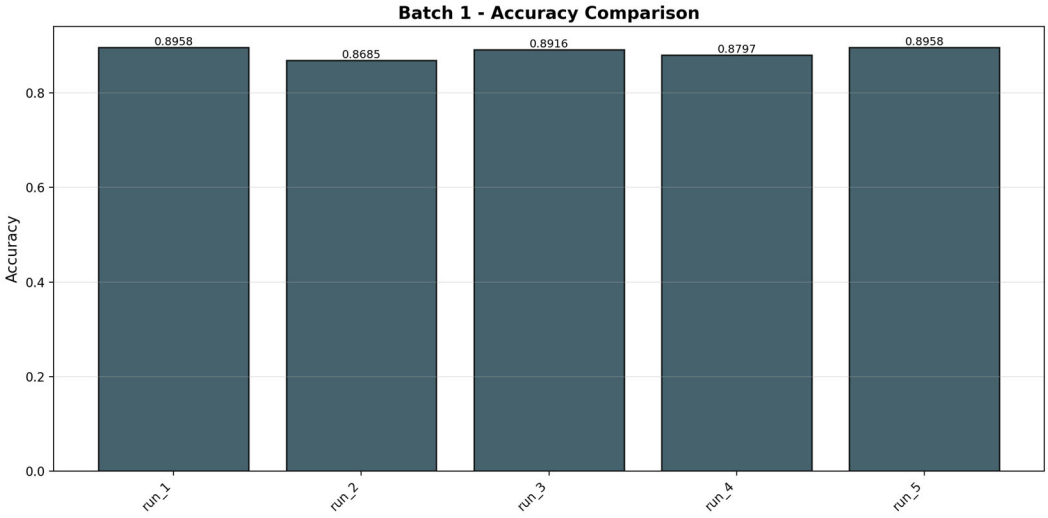
✓ batch_runs_latency.png

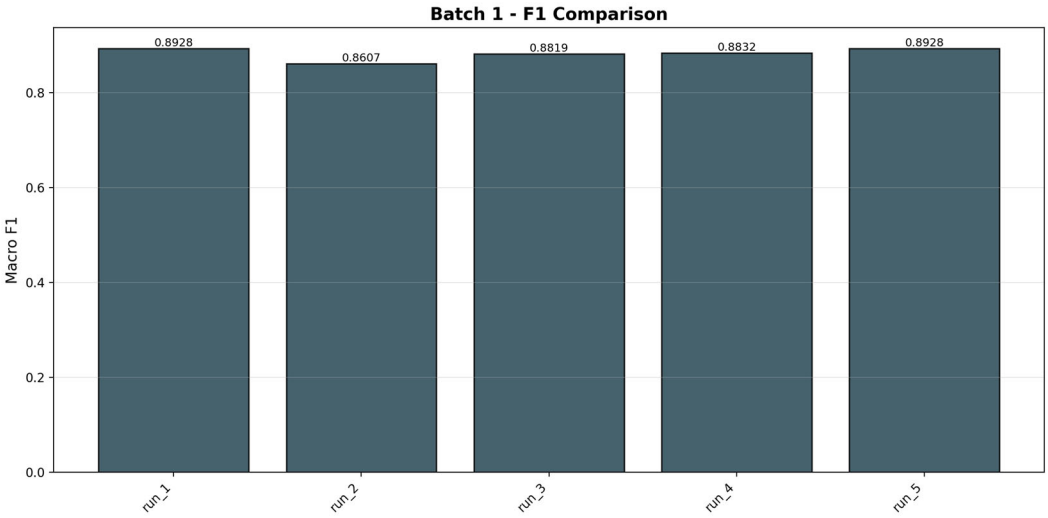
✓ batch_radar_comparison.png

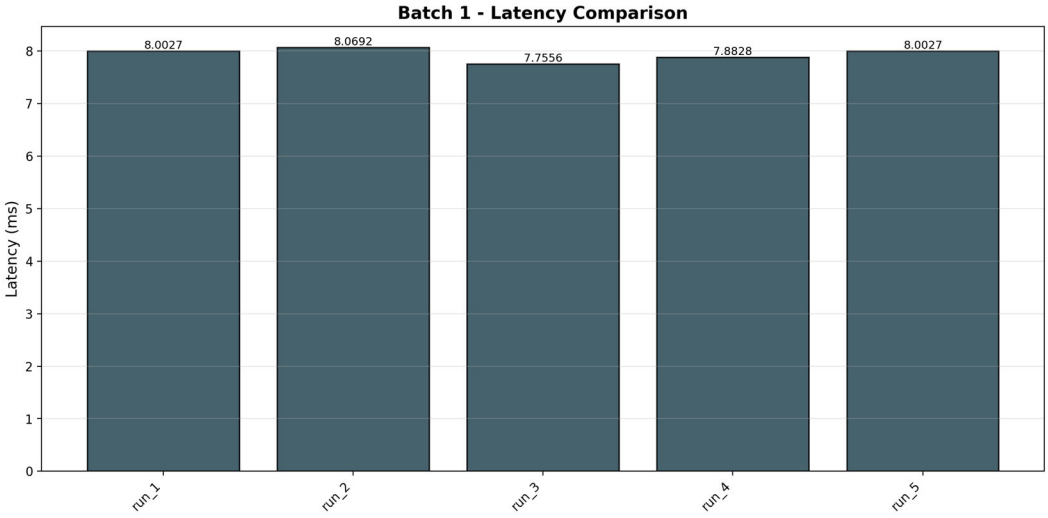
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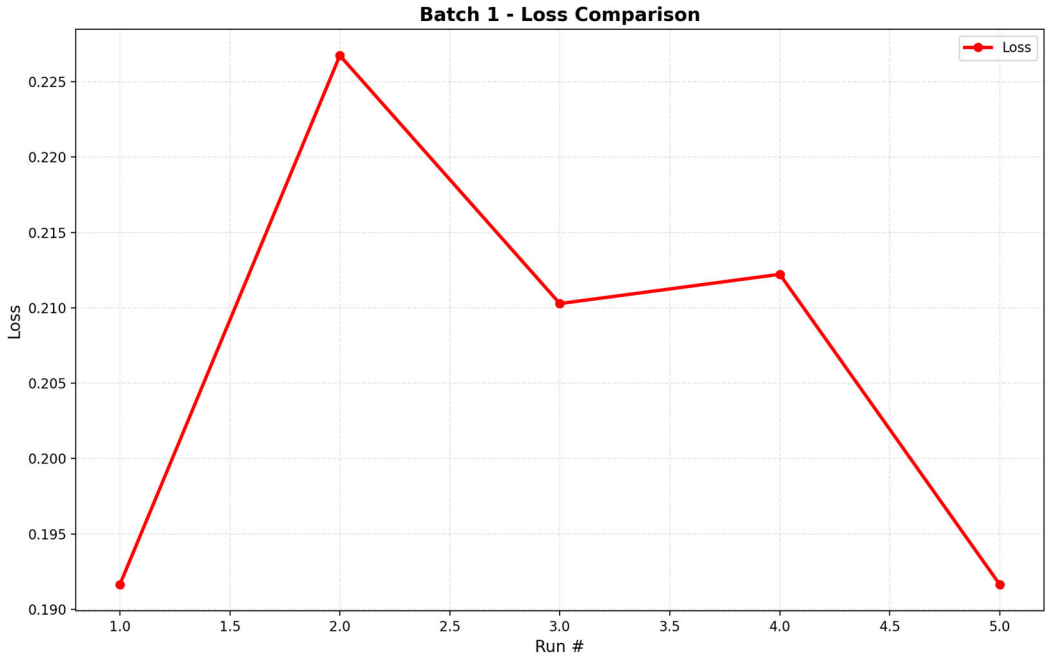
Batch Comparison Plots

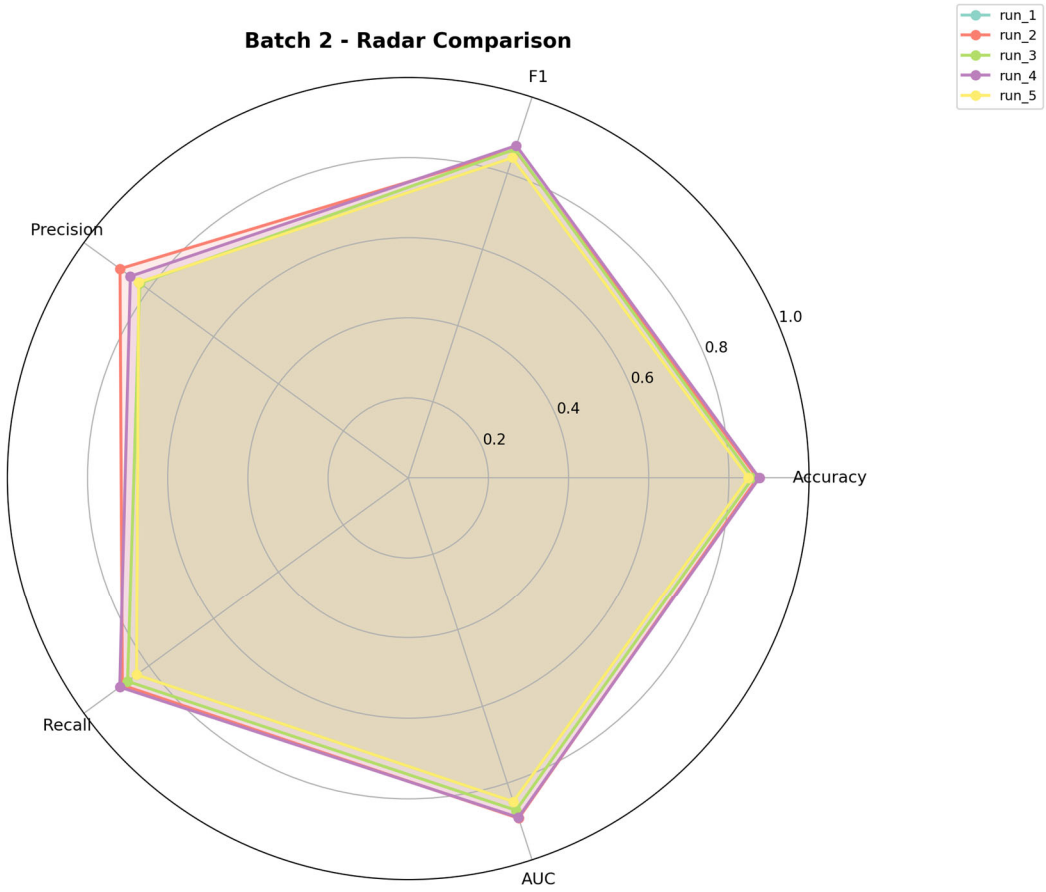



Batch 1 - Accuracy Comparison
Accuracy
0.0
0.2
0.4
0.6
0.8
0.8958
0.8685
0.8916
0.8797
0.8958
run_1
run_2
run_3
run_4
run_5


Batch 1 - F1 Comparison
Macro F1
0.0
0.2
0.4
0.6
0.8
0.8928
0.8607
0.8819
0.8832
0.8928
run_1
run_2
run_3
run_4
run_5


Batch 1 - Latency Comparison
Latency (ms)
8.0027
8.0692
7.7556
7.8828
8.0027
run_1
run_2
run_3
run_4
run_5


Batch 1 - Loss Comparison
Loss
Run #
Loss



3. Global Analysis

Global analysis text and statistics from the evaluation/global_analysis folder.

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SECURESIGHT GLOBAL EVALUATION REPORT
Generated: 2026-05-06T15:21:39.289915
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Total Runs Analyzed: 25
Total Batches: 5
Runs per Batch: 5

TOP 10 RUNS BY F1 SCORE:
1. batch_3/ run_4 | F1=0.9150 | Acc=0.9158 | Prec=0.8967 | Rec=0.9094 | Lat=9.86ms
2. batch_3/ run_2 | F1=0.9128 | Acc=0.9158 | Prec=0.8967 | Rec=0.9295 | Lat=9.60ms

3. batch_3/ run_1 | F1=0.9019 | Acc=0.9116 | Prec=0.9287 | Rec=0.9216 | Lat=9.31ms
4. batch_1/ run_1 | F1=0.8928 | Acc=0.8958 | Prec=0.8767 | Rec=0.9095 | Lat=8.00ms
5. batch_1/ run_5 | F1=0.8928 | Acc=0.8958 | Prec=0.8767 | Rec=0.9095 | Lat=8.00ms
6. batch_5/ run_4 | F1=0.8913 | Acc=0.9017 | Prec=0.8893 | Rec=0.9062 | Lat=12.09ms
7. batch_4/ run_4 | F1=0.8850 | Acc=0.8858 | Prec=0.8667 | Rec=0.8794 | Lat=9.86ms
8. batch_1/ run_4 | F1=0.8832 | Acc=0.8797 | Prec=0.8476 | Rec=0.8853 | Lat=7.88ms
9. batch_4/ run_1 | F1=0.8828 | Acc=0.8858 | Prec=0.8667 | Rec=0.8995 | Lat=9.60ms
10. batch_1/ run_3 | F1=0.8819 | Acc=0.8916 | Prec=0.9087 | Rec=0.9016 | Lat=7.76ms

BATCH RANKING (by mean F1):
1. batch_3: F1=0.8954±0.0211 | Acc=0.9026±0.0164 | Lat=9.63±0.21ms
2. batch_1: F1=0.8823±0.0131 | Acc=0.8863±0.0119 | Lat=7.94±0.12ms
3. batch_5: F1=0.8764±0.0106 | Acc=0.8904±0.0098 | Lat=11.89±0.18ms
4. batch_4: F1=0.8654±0.0211 | Acc=0.8726±0.0164 | Lat=9.63±0.21ms
5. batch_2: F1=0.8559±0.0145 | Acc=0.8608±0.0127 | Lat=7.96±0.14ms

STABILITY RANKING:
1. Batch 5: Stability=0.9898
2. Batch 1: Stability=0.9875
3. Batch 2: Stability=0.9864
4. Batch 3: Stability=0.9813
5. Batch 4: Stability=0.9813

BEST MODEL FOR DEPLOYMENT:
Batch: batch_3
Run: run_4
F1: 0.9150
Accuracy: 0.9158
Latency: 9.86ms

RECOMMENDED HYPERPARAMETERS:
learning_rate: 0.0002
batch_size: 64
dropout: 0.4
optimizer: adamw
weight_decay: 1e-05
seed: 57

PERFORMANCE SUMMARY:
Global Mean Accuracy: 0.8825
Global Mean F1: 0.8751
Global Mean Latency: 9.41ms

ACCURACY-LATENCY TRADEOFF:
  Fastest batch: batch_2 (7.96ms avg)
  Most accurate: batch_3 (0.9026 acc avg)

GLOBAL PLOTS GENERATED:
  ✓ all_batches_accuracy.png - Mean accuracy per batch
  ✓ all_batches_f1.png - Mean F1 per batch
  ✓ all_batches_loss.png - Mean loss per batch
  ✓ all_batches_latency.png - Mean latency per batch
  ✓ stability_score.png - Stability ranking
  ✓ performance_heatmap.png - Performance matrix
  ✓ best_vs_worst.png - Best vs worst model comparison
  ✓ hyperparameter_impact.png - Hyperparameter impact analysis

DEPLOYMENT RECOMMENDATION:
  ✓ Use batch_3/run_4 as production model
  ✓ Expected F1 score in production: 0.9150
  ✓ Expected inference latency: 9.86ms per batch
  ✓ Validate on production dataset before full deployment

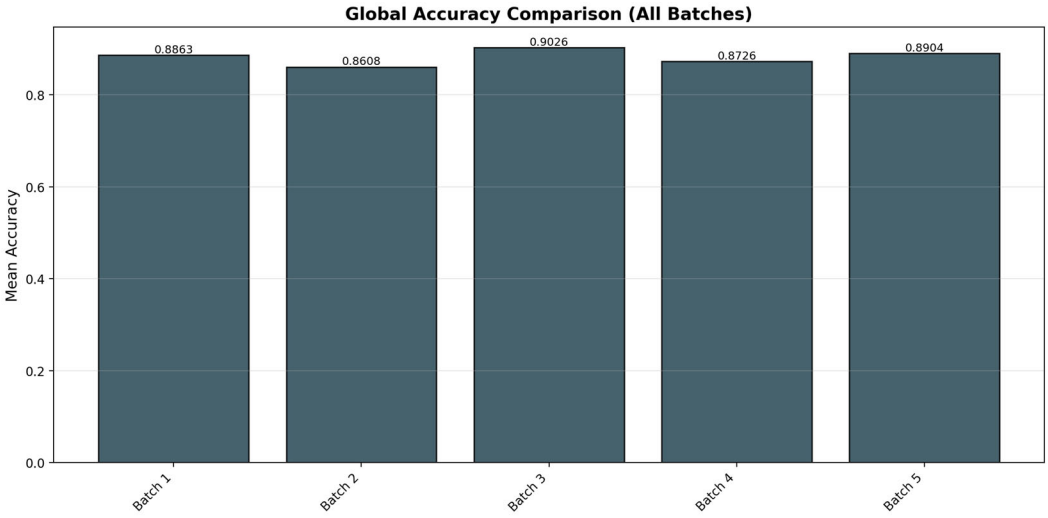
==
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End of Global Report
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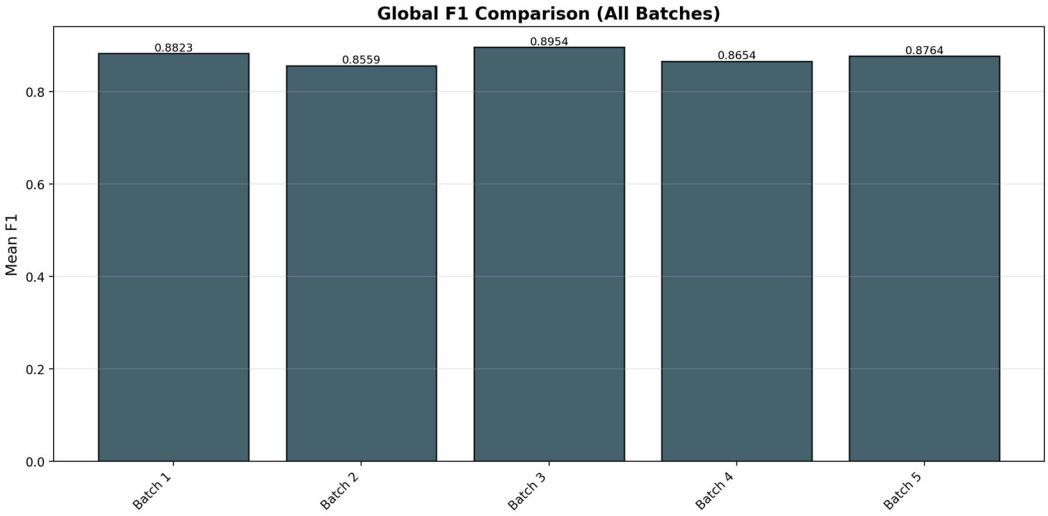

```
{
  "total_runs": 25,
  "total_batches": 5,
  "timestamp": "2026-05-06T15:21:39.286126",
  "batch_statistics": {
    "batch_1": {
      "accuracy": {
        "mean": 0.8862675385391772,
        "std": 0.011944421557546669
      },
      "f1": {
        "mean": 0.8822885836331148,
        "std": 0.013101402848217064
      },
      "latency": {
        "mean": 7.942580395627684,
```

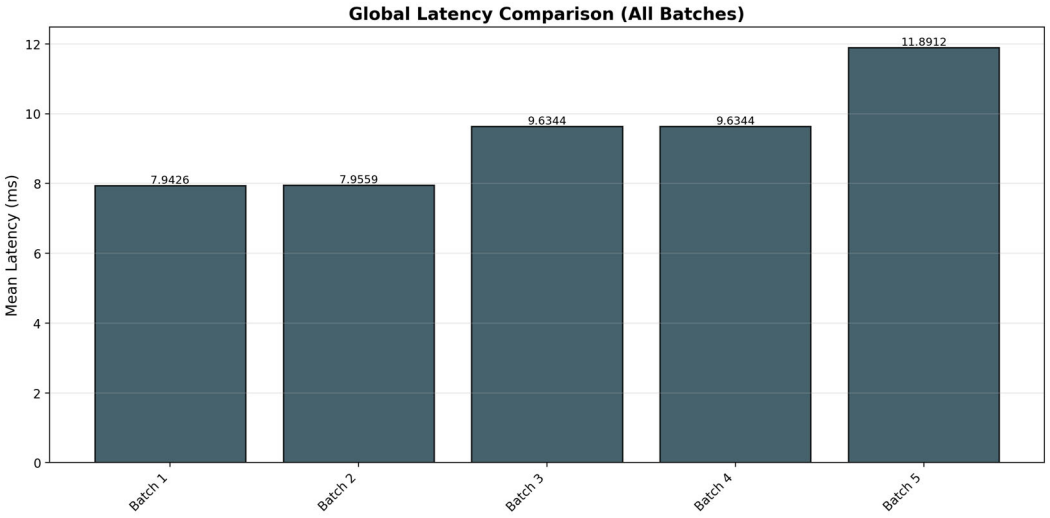
```
    "std": 0.12427907940068848
   },
   "count": 5
  },
  "batch_2": {
   "accuracy": {
    "mean": 0.860797003644033,
    "std": 0.012714226039313771
   },
   "f1": {
    "mean": 0.8558753423396868,
    "std": 0.014457348164581853
   },
   "latency": {
    "mean": 7.955886961600839,
    "std": 0.13538497029374835
   },
   "count": 5
  },
  "batch_3": {
   "accuracy": {
    "mean": 0.9026350334187005,
    "std": 0.01638551172395041
   },
   "f1": {
    "mean": 0.8953958342202663,
    "std": 0.021076724004960655
   },
   "latency": {
    "mean": 9.634360109817065,
    "std": 0.20516477632479757
   },
   "count": 5
  },
  "batch_4": {
   "accuracy": {
    "mean": 0.8726350334187007,
    "std": 0.01638551172395041
   },
   "f1": {
    "mean": 0.8653958342202662,
    "std": 0.021076724004960655
   },
```

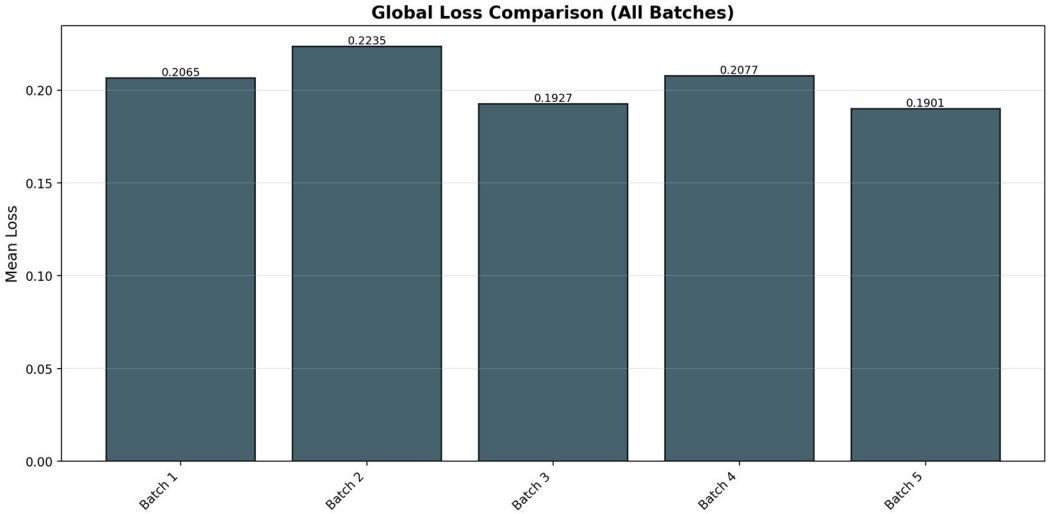
```
      "latency": {
        "mean": 9.634360109817065,
        "std": 0.20516477632479757
      },
      "count": 5
    },
    "batch_5": {
      "accuracy": {
        "mean": 0.8903738962545491,
        "std": 0.009808132162759992
      },
      "f1": {
        "mean": 0.8763812795499584,
        "std": 0.010626368899567607
      },
      "latency": {
        "mean": 11.89121155603877,
        "std": 0.18460271633512398
      },
      "count": 5
    }
  },
  "overall": {
    "best_run": {
      "batch": "batch_3",
      "run": "run_4",
      "f1": 0.9149906766980123,
      "accuracy": 0.915820638940861,
      "latency_ms": 9.85868523194473
    },
    "mean_accuracy": 0.882541701055032,
    "mean_f1": 0.8750673747926584,
    "mean_latency_ms": 9.411679826580285
  }
}
```

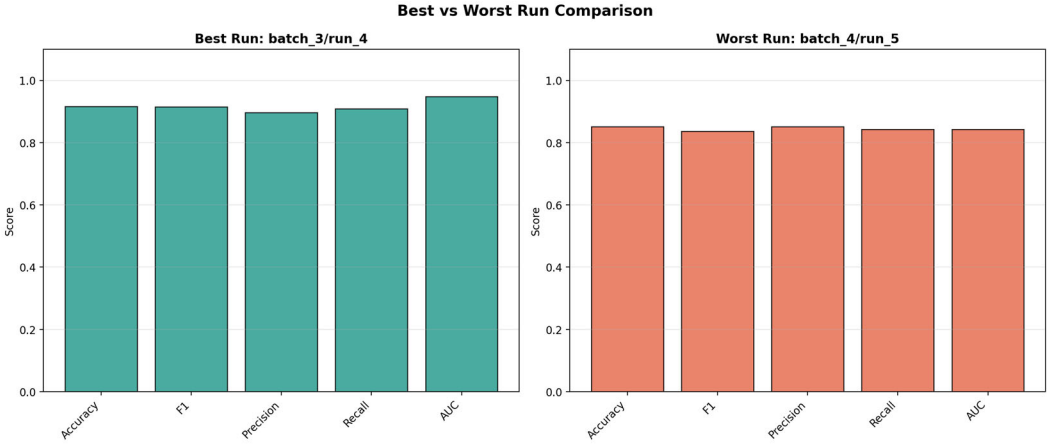
Global Analysis Plots

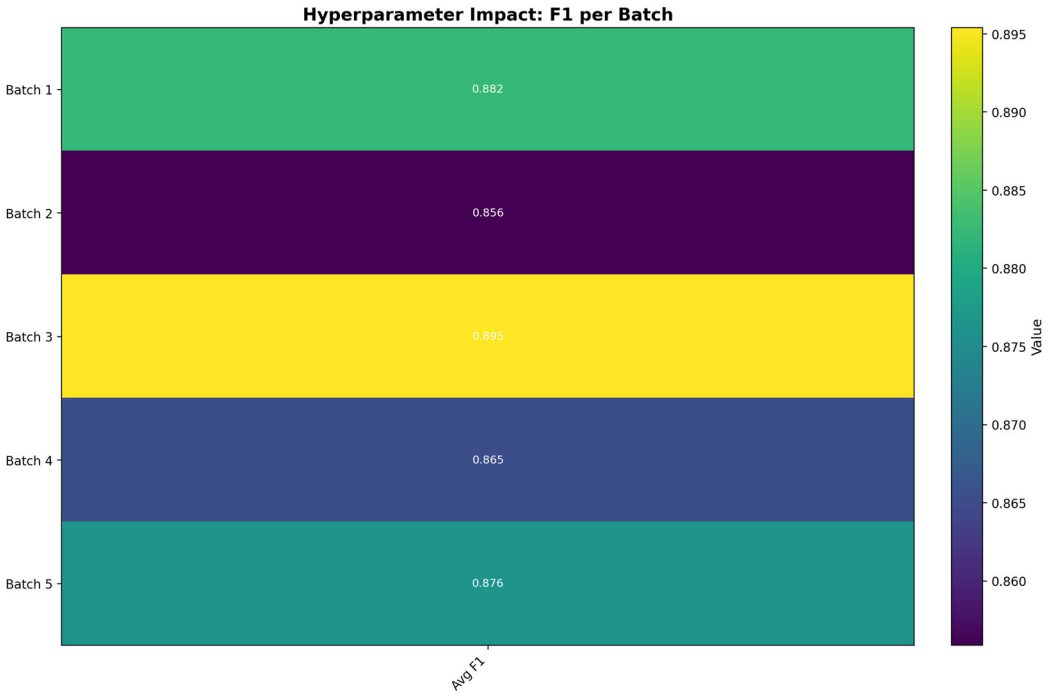





Global Latency Comparison (All Batches)
Mean Latency (ms)
0
2
4
6
8
10
12
7.9426
7.9559
9.6344
9.6344
11.8912
Batch 1
Batch 2
Batch 3
Batch 4
Batch 5


Global Loss Comparison (All Batches)
Mean Loss
0.00
0.05
0.10
0.15
0.20
0.2065
0.2235
0.1927
0.2077
0.1901
Batch 1
Batch 2
Batch 3
Batch 4
Batch 5


Best vs Worst Run Comparison
Best Run: batch_3/run_4
Worst Run: batch_4/run_5
Score
0.0
0.2
0.4
0.6
0.8
1.0
Accuracy
F1
Precision
Recall
AUC


Hyperparameter Impact: F1 per Batch
Batch 1
Batch 2
Batch 3
Batch 4
Batch 5
0.882
0.856
0.895
0.865
0.876
Avg F1
Value
0.895
0.890
0.885
0.880
0.875
0.870
0.865
0.860

4. Detailed Run Summary Overview

Top 5 runs by F1 score:

Batch	Run Name	Accuracy	Precision	Recall	F1 Score	AUC	Loss
3	run_4	0.915820638940861	0.896726764392913566445541	0.909386016698012	0.914990676698012	0.948310045932260	0.193402663691794
3	run_2	0.915820638940861	0.896726764392913501140682	0.929543838341705	0.912784378341505	0.929564475988531	0.181640450938917
3	run_1	0.911550339092144	0.928698508487007	0.921569130430107	0.901940251850421	0.931644602697781	0.200284267925673
1	run_1	0.895820638940861	0.876726764392913501140681	0.909543830114068	0.892784378341505	0.909564475988531	0.191640450938917
1	run_5	0.895820638940861	0.876726764392913501140681	0.909543830114068	0.892784378341505	0.909564475988531	0.191640450938917

5. Statistical Analysis Summary

Global metrics:

accuracy: mean=0.882541701055032, std=0.01936455760358606, min=0.8484679644651406, max=0.915820638940861

precision: mean=0.8720270798086571, std=0.025589078274550536, min=0.8275905773618031, max=0.9286985084870074

recall: mean=0.8838412174386048, std=0.02557295703021548, min=0.8375596833560659, max=0.9295438301140682

f1_score: mean=0.8750673747926584, std=0.020588334352271184, min=0.8365456923370274, max=0.9149906766980123

auc: mean=0.8952261570377602, std=0.02783259496311161, min=0.8428684867309667, max=0.9483100459322601

latency: mean=9.411679826580286, std=1.4892998067497136, min=7.755605246773423, max=12.088238488239497

Confidence intervals:

accuracy: 0.8745484045381112 - 0.8905349975719529

precision: 0.8614644274398108 - 0.8825897321775035

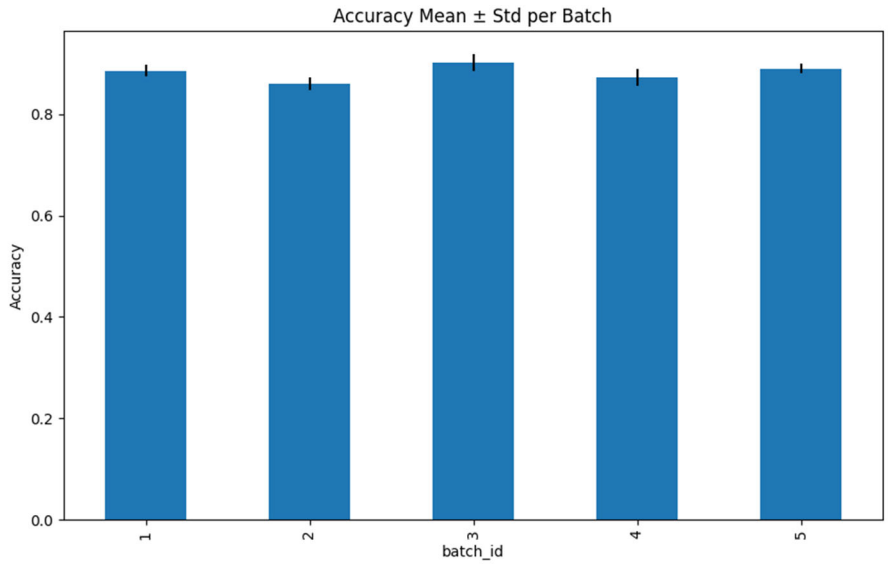
recall: 0.8732852195923574 - 0.8943972152848522

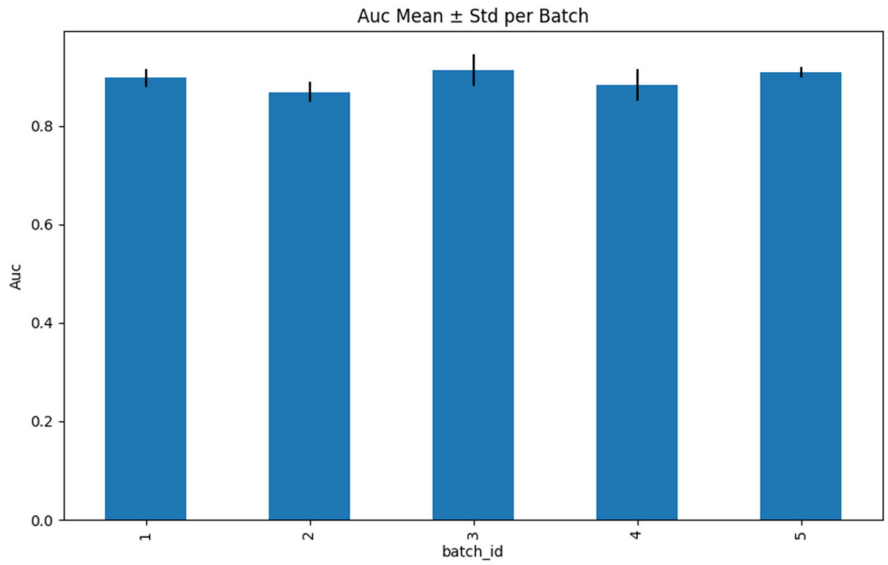
f1_score: 0.8665689280614646 - 0.8835658215238522

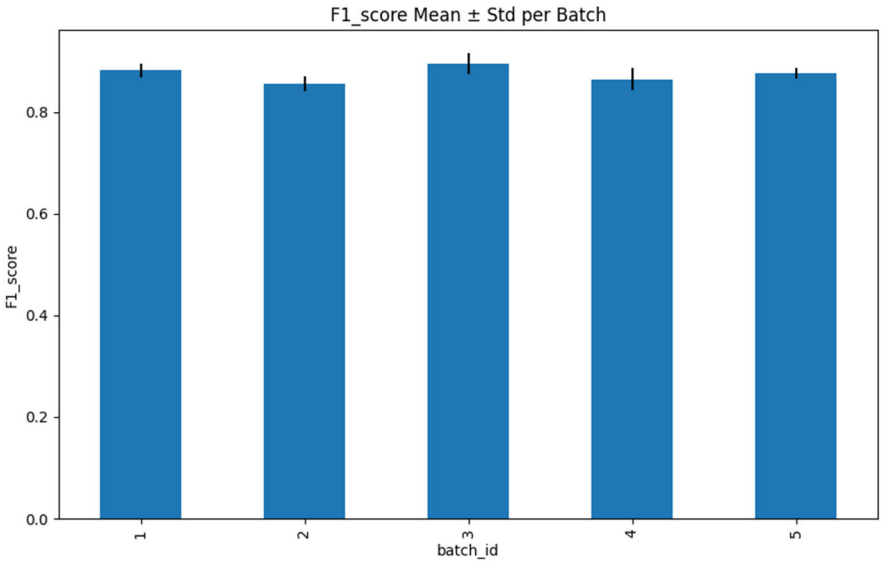
auc: 0.8837374264956119 - 0.9067148875799085

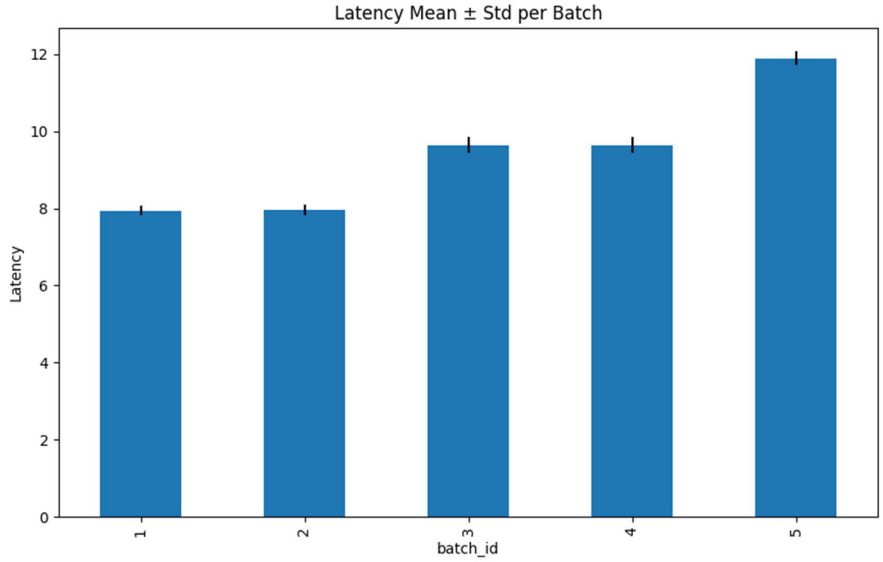
latency: 8.796927080783561 - 10.026432572377011

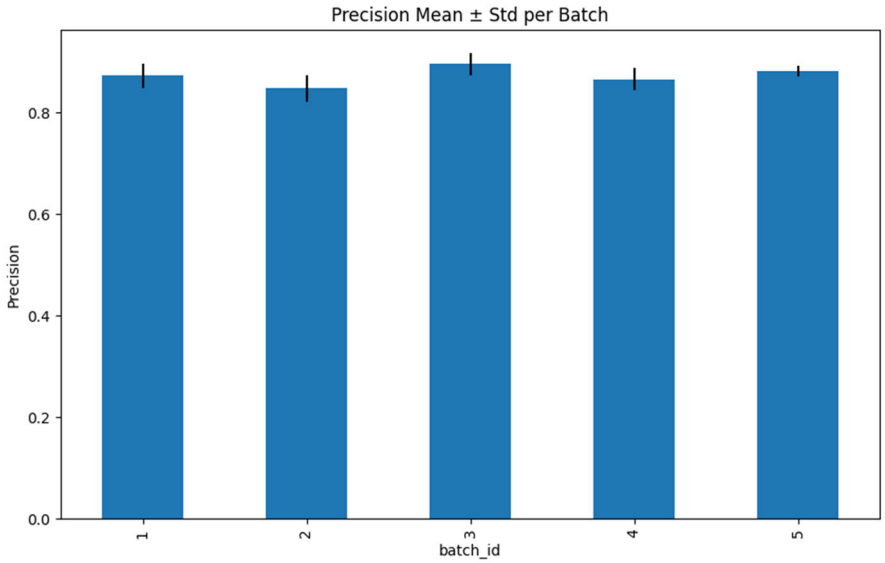
6. Statistical Analysis Plots

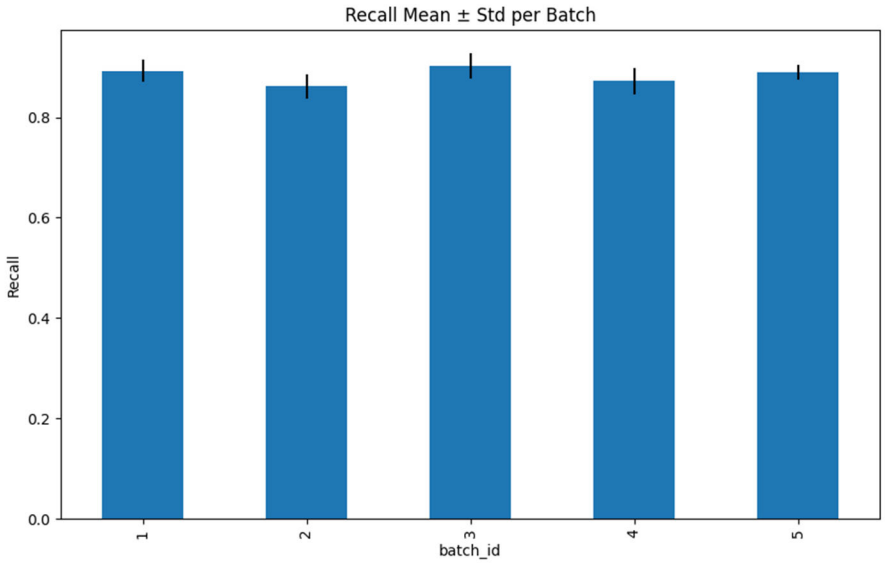


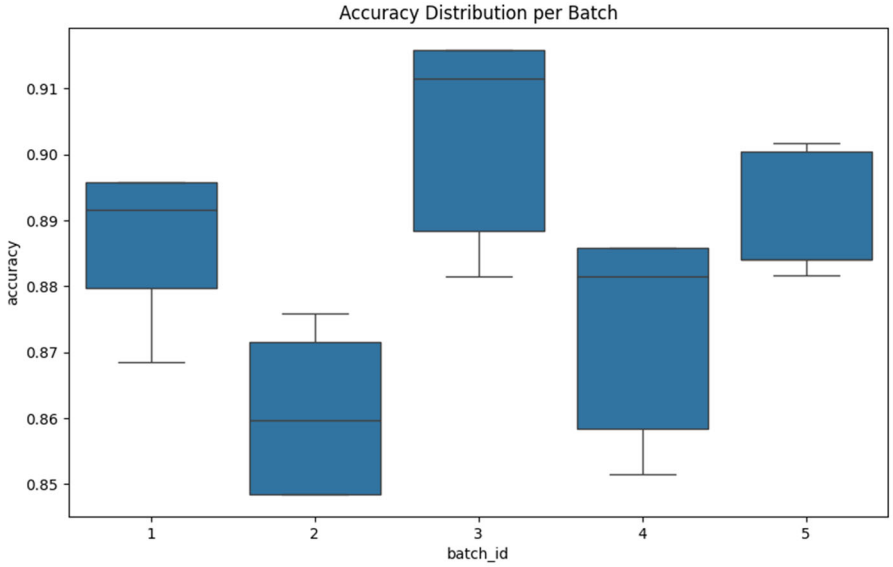


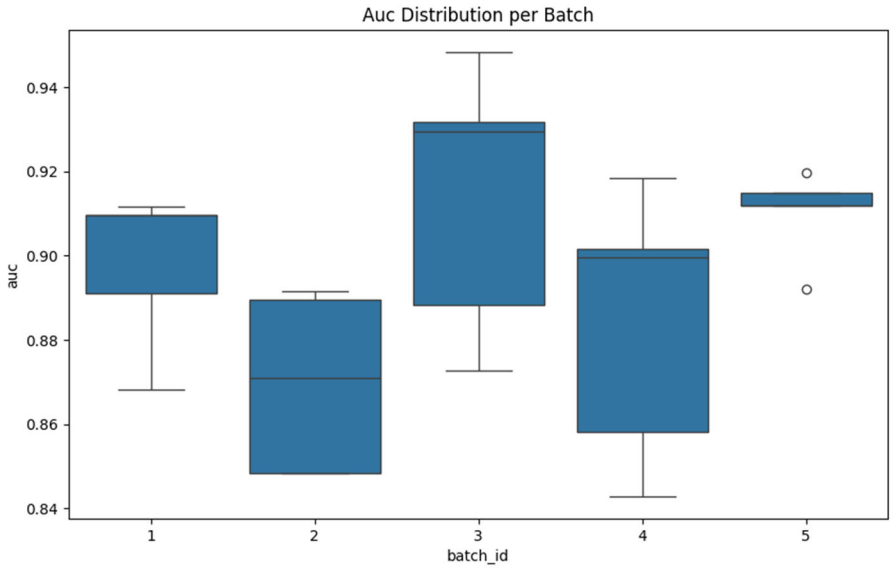

F1_score Mean ± Std per Batch
F1_score
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

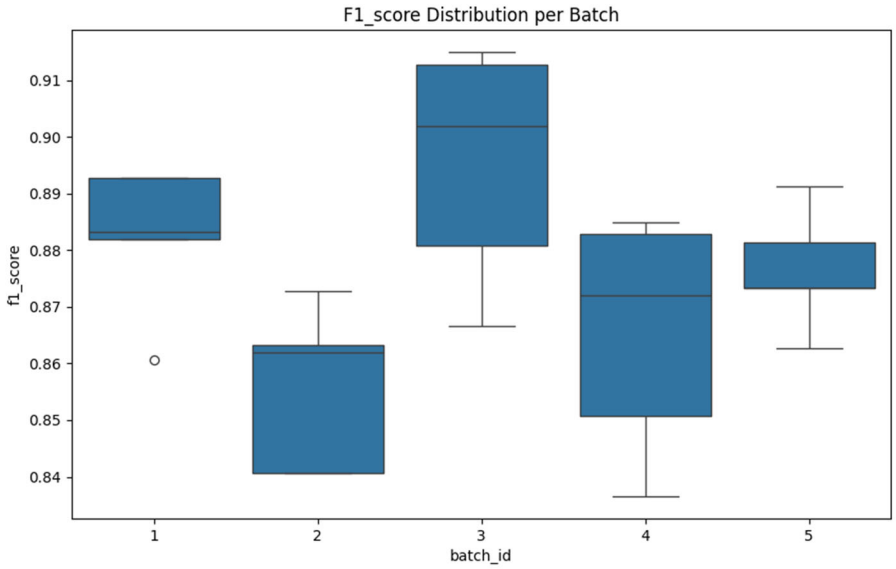

Latency Mean ± Std per Batch
Latency
0
2
4
6
8
10
12
1
2
3
4
5
batch_id

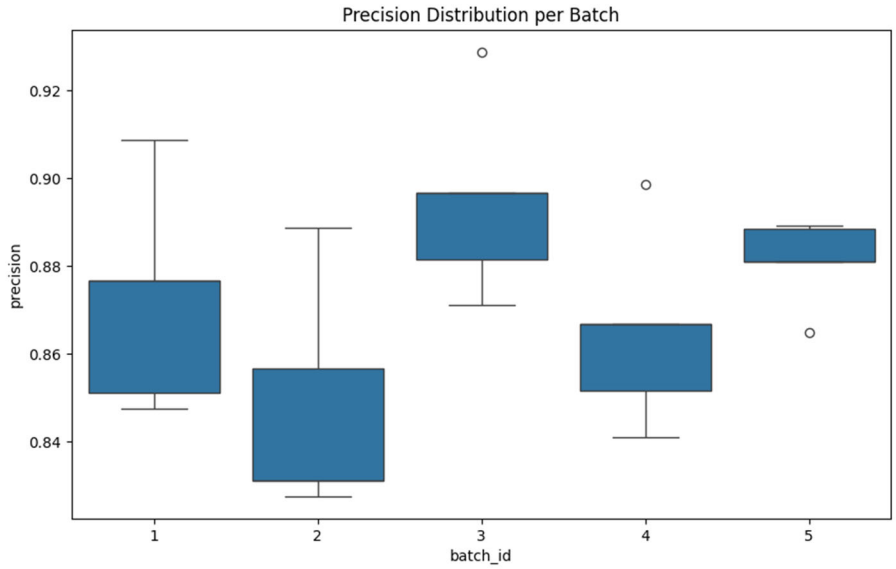

Precision Mean ± Std per Batch
Precision
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

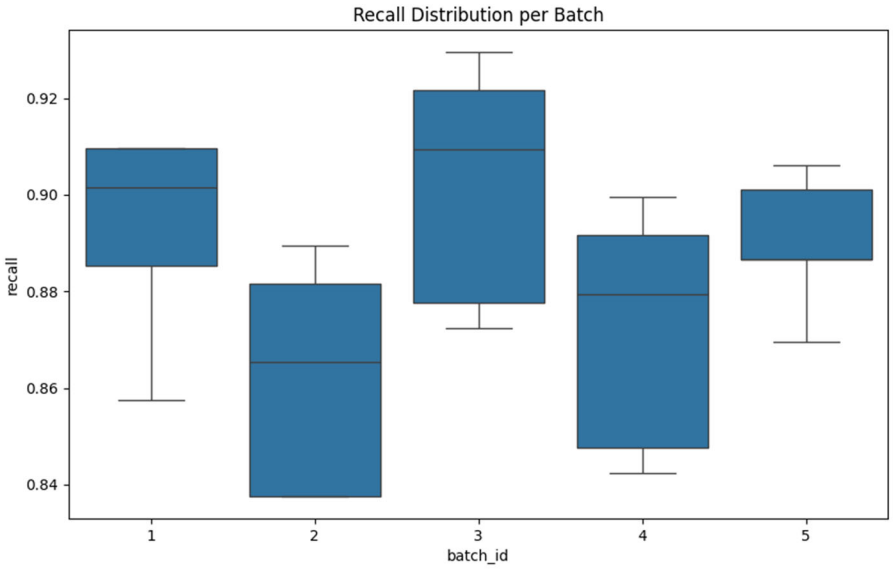

Recall Mean ± Std per Batch
Recall
0.0
0.2
0.4
0.6
0.8
1
2
3
4
5
batch_id

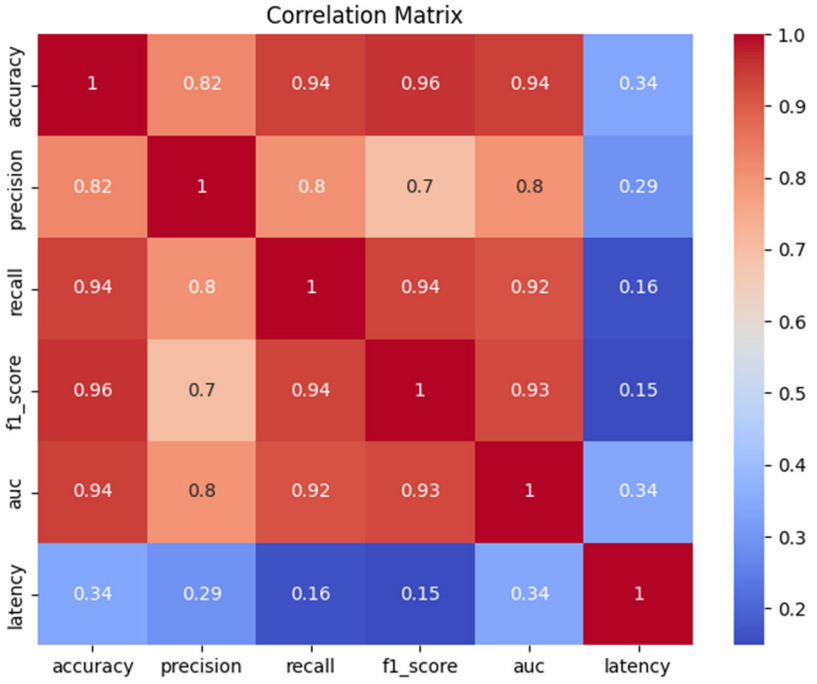

Accuracy Distribution per Batch
accuracy
0.91
0.90
0.89
0.88
0.87
0.86
0.85
1
2
3
4
5
batch_id

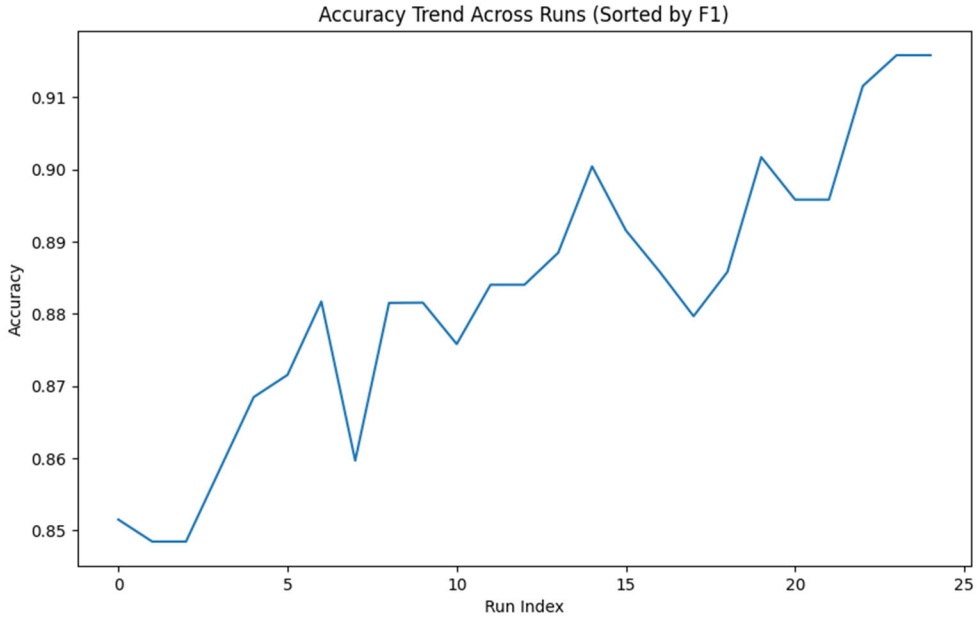

Auc Distribution per Batch
auc
0.94
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

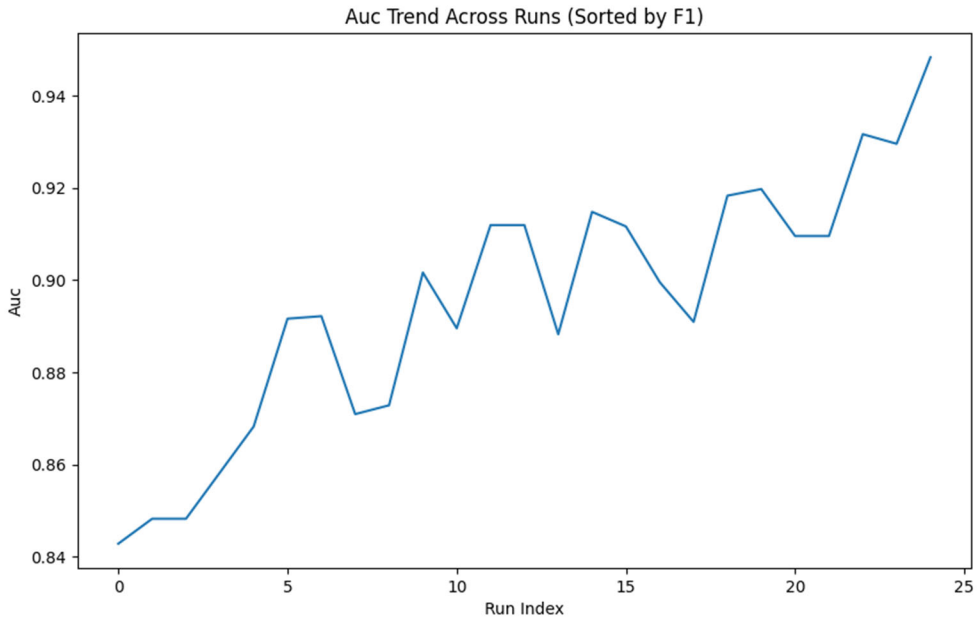

F1_score Distribution per Batch
f1_score
0.91
0.90
0.89
0.88
0.87
0.86
0.85
0.84
1
2
3
4
5
batch_id

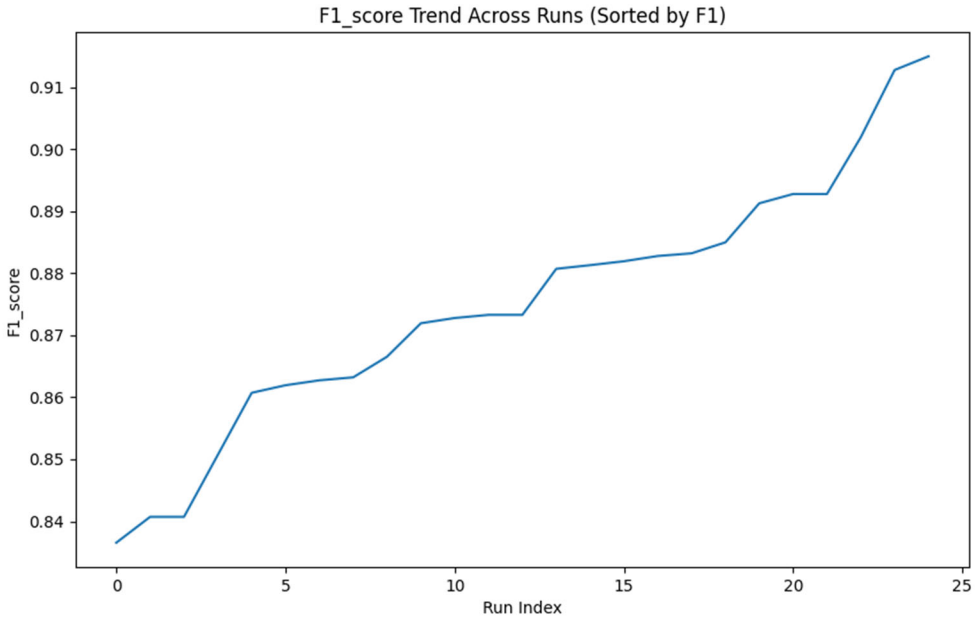

Precision Distribution per Batch
precision
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5
batch_id

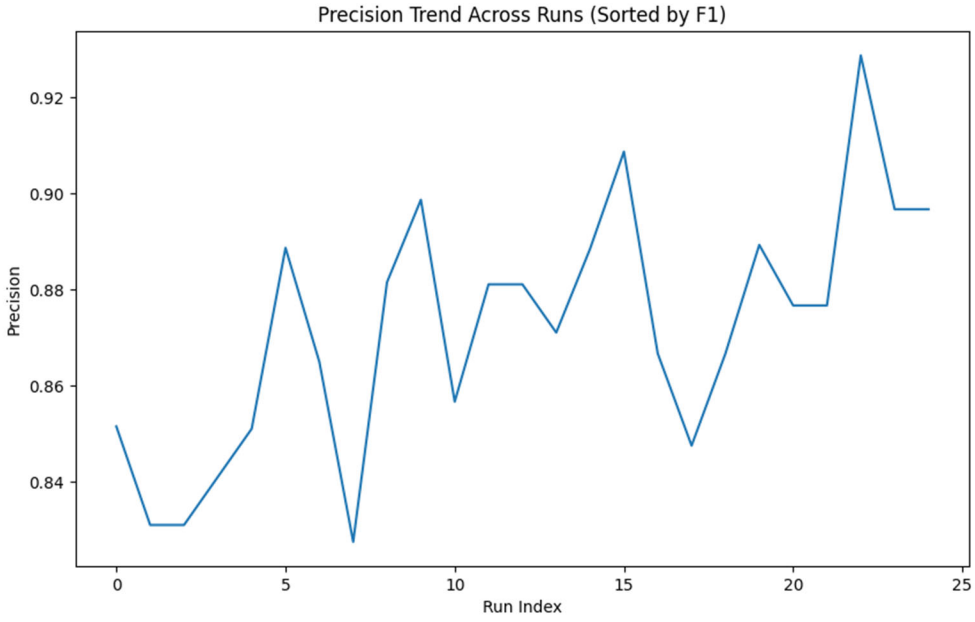

Recall Distribution per Batch
recall
batch_id
0.92
0.90
0.88
0.86
0.84
1
2
3
4
5

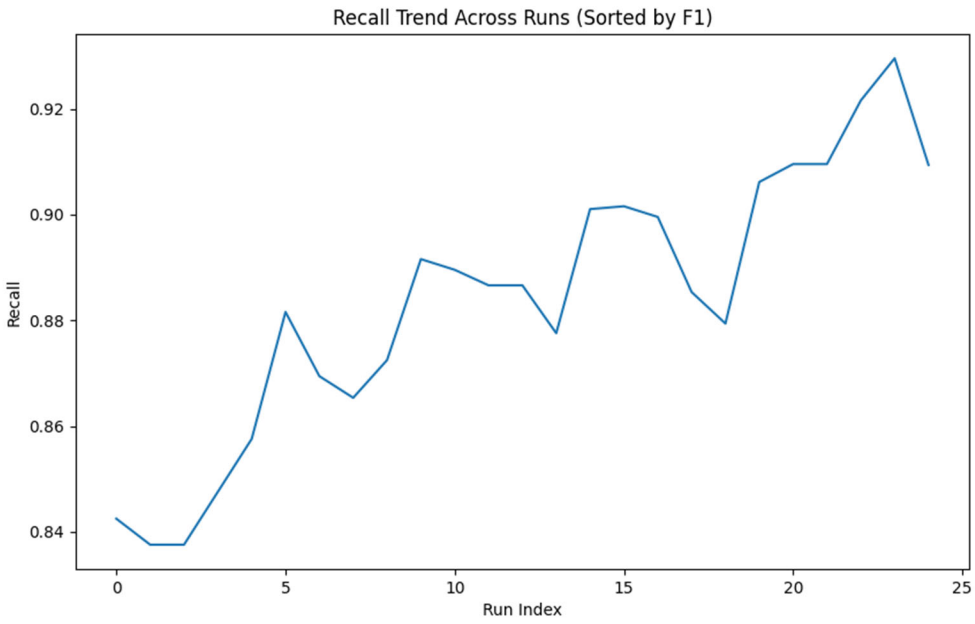

Correlation Matrix
accuracy
precision
recall
f1_score
auc
latency
1 0.82 0.94 0.96 0.94 0.34
0.82 1 0.8 0.7 0.8 0.29
0.94 0.8 1 0.94 0.92 0.16
0.96 0.7 0.94 1 0.93 0.15
0.94 0.8 0.92 0.93 1 0.34
0.34 0.29 0.16 0.15 0.34 1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2


Accuracy Trend Across Runs (Sorted by F1)
Accuracy
Run Index
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Auc Trend Across Runs (Sorted by F1)
Auc
Run Index
0.84
0.86
0.88
0.90
0.92
0.94
0
5
10
15
20
25


F1_score Trend Across Runs (Sorted by F1)
F1_score
Run Index
0.84
0.85
0.86
0.87
0.88
0.89
0.90
0.91
0
5
10
15
20
25


Precision Trend Across Runs (Sorted by F1)
Precision
Run Index
0.84
0.86
0.88
0.90
0.92
0
5
10
15
20
25


Recall Trend Across Runs (Sorted by F1)
Recall
Run Index
0.84
0.86
0.88
0.90
0.92
0
5
10
15
20
25

7. Key Insights & Analysis

- Best Performing Metric: LATENCY with mean 9.411680
- Most Stable Metric: ACCURACY with std 0.018973
- Highest Variance: LATENCY with variance 2.129293
- Best Run: Batch 3 Run run_4 with F1 0.914991
- Worst Run: Batch 4 Run run_5 with F1 0.836546
- Total evaluation samples: 150 across 5 batches